Divya Upadhyay

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AI Software Engineer with 5+ years of experience architecting **GenAI (RAG) systems** and real-time **Computer Vision pipelines**. Specialist in productionizing **AI models**, focusing on rigorous **CI/CD** workflows, automated testing, and **AWS** infrastructure to ensure high availability. Expert in optimizing inference costs and latency for enterprise-scale applications.

EXPERIENCE (5+ years)

AI Software Engineer | Golok Global INC, USA **June 2025 – Present**

- Architected an AI-driven **customer support automation platform**, cutting query resolution time by **11%** using **Gemma** with **RAG** and **context engineering**; built intent classification, confidence-based routing, and human-in-the-loop escalation for reliability.
- Designed and deployed a real-time **AI-powered conversational system** for a ride-hailing platform, integrating **LLM** orchestration, **RAG**-based knowledge grounding, and workflow automation for booking assistance and issue resolution at scale.
- Built and scaled a low-latency payment processing API using FastAPI with Redis-based caching and idempotency keys, **handling 500+ concurrent requests with sub-80ms P95 internal processing time before async handoff to Stripe.**
- Implemented an automated **MLOps** pipeline integrating **GitHub, Jenkins**, **Docker-based model packaging**, **AWS ECS**, **CI/CD validation**, and staged deployments, reducing model release latency by **95%** while enforcing reproducibility and rollback safety.
- **Led** backend system design for a scalable travel platform, owning end-to-end service architecture, **API contracts**, and **AWS cloud deployment**; integrated **React** clients with **Java Spring Boot microservices**.

Data Scientist | Health Innovators INC, USA **August 2024 – June 2025**

- Designed and deployed **LLM-powered** conversational pipelines for telehealth workflows using **OpenAI APIs** with structured **prompting**, **function calling**, and **RAG** over clinical FAQs, enabling automated symptom screening and patient triage; evaluated performance achieving an **18% increase in intent classification F1 scores and a 25% boost in resolution rates.**
- Architected a **multi-agent** clinical decision-support system using **LLM orchestration** (router + specialist agents, including a Radiology Agent) with report embedding, cross-agent context sharing, and chain-of-thought–constrained reasoning, improving clinician task completion time by **22%** and reducing manual report lookup latency by **35%**.
- Implemented scalable ML model serving via **FastAPI** with async I/O, **Dockerized** deployments, and **Kubernetes HPA** for auto-scaling; instrumented **Prometheus/Grafana for latency and error-rate monitoring**, achieving **99.9% service uptime** and **sub-120ms P95** inference latency in production.
- Built a clinical NLP information extraction pipeline using Hugging Face Transformers (**fine-tuned BioClinicalBERT / PubMedBERT**) with **NER** and entity normalization to map unstructured notes and pathology reports to structured **FHIR-aligned schemas**, improving extraction **F1-score to 0.86** and increasing chart processing throughput by **60%**.
- Optimized analytical **SQL pipelines** supporting **daily ingestion of ~50GB of patient interaction data**, refactoring joins and indexing strategies to reduce query latency by **70%** and meet healthcare reporting **SLAs.**

Software Engineer (DevOps) | Nissan Digital India LLP, India | 10 months Co-op **September 2018 – June 2022**

- Engineered **Next-Gen Delivery platform;** fully automated CI/CD pipelines using **Jenkins, GitHub, SonarQube, Veracode & Nexus** integrating automated testing, artifact versioning, and deployment gates to reduce release cycle time by **40%.**
- Automated infrastructure provisioning and configuration management using **Ansible** & **AWS CloudFormation**, enabling zero-touch environment spin-up for dev, staging, and production with reproducible, immutable infrastructure patterns.
- Implemented observability and alerting across critical services using **Prometheus, Grafana, and the ELK stack**, establishing SLO-driven monitoring and proactive alerting that reduced incident response time and improved platform availability.
- Developed **computer vision**–based **automated visual inspection pipelines** for vehicle body panels using image preprocessing and defect detection models, replacing manual spot-check workflows and improving defect detection efficiency by **25%**.

SKILLS

Machine Learning (Agentic & Production Focus): Agentic Orchestration (LangGraph), DSPy (Programmatic Prompting), LLM Evaluation, Efficient Inference (vLLM), Parameter-Efficient Fine-Tuning (QLoRA), TensorFlow, PyTorch, Scikit-learn
Infrastructure (LLMOps & Platform Focus): LLMOps, LangSmith (AI Observability), AWS (SageMaker, Lambda, S3, EC2), Docker, Kubernetes, GitHub Actions (CI/CD), MLflow, Prometheus, Grafana, FastAPI, REST APIs, Git
Data Engineering (Context & Retrieval Focus): GraphRAG (Knowledge Graphs), Apache Iceberg (Open Table Formats), Advanced Vector Search, Real-time Context (Apache Flink), PySpark, Apache Airflow, PostgreSQL, Pandas, NumPy.

PROJECTS

DeepNucleiNet Nuclei Instance Segmentation **March 2024 – April 2024**

- Performed instance nuclei segmentation (37,333 nuclei images) by fine-tuning **YOLOv8, SAM,** and **Detectron2** frameworks.
- Achieved **mAP of 0.49 with Detectron2,** outperforming **SAM (IoU: 0.21)** and **YOLOv8,** validating architectural trade-offs for histopathology image analysis.

CodeLings: Natural Language to Bash Script Translator **March 2023 – April 2023**

- Fine-tuned a **codet5p-220m-bimodal model** for translating **natural language to Bash commands,** implementing a **seq2seq pipeline** with custom dataset processing and achieving a **BLEU score of 0.3939 and precision of 0.55.**
- Deployed on Hugging Face to enhance command-line accessibility for users and allow integration for developers.

EDUCATION

- **MS in Computer and Information Sciences |** University of Florida **[May 2024] | GPA: 3.90 / 4.00**
- **B-Tech in Computer Science and Engineering |** KIIT University, India **[May 2019] | GPA: 8.98 / 10.00**